

Lesson 17: Alternating Series

Rubric for Evaluating Series Convergence

Criterion	Exemplary (3)	Proficient (2)	Developing (1)
Identification	Correct identifying the series as alternating and isolating the non-negative sequence a_n .	Correct identifier, minor errors in isolating a_n .	Fails to recognize as alternating or incorrectly identifies a_n .
Decreasing Sequence Test	Demonstrates $a_{n+1} \leq a_n$ for all n (or eventually) using derivative/algebra.	Mentions a_n decreases but lacks formal proof or shows it for only a few terms.	Fails to check if a_n is decreasing.
Limit Test	Clearly calculates $\lim_{n \rightarrow \infty} b_n = 0$.	Evaluates limit but with minor calculation errors.	Limit is not zero or not calculated.
Conclusion	Correctly concludes convergence/divergence based on AST, acknowledging test limitations.	Concludes convergence but with weak justification.	Incorrect conclusion.